The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 November 2001.

INCH-POUND

MIL-PRF-19500/421F
25 August 2001
SUPERSEDING
MIL-PRF-19500/421E
2 April 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DUAL TRANSISTOR, UNITIZED, NPN/PNP, COMPLEMENTARY, SILICON TYPES 2N3838, 2N4854, AND 2N4854U JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for unitized, dual transistors which contain a pair of electrically isolated complementary NPN and PNP silicon triode transistors in one package. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Polarity designation. Voltages and currents of limits and test conditions shown herein apply to the NPN triode. For the PNP triode, the values are the same, but the polarity designations shall be the opposite.

1.3 Physical dimensions. See figures 1 (6 lead flatpak), 2 (similar to T0-77), and 3 (surface mount).

1.4 Maximum ratings.

P_T at T_A = +25°C				P_T at T_C = +25°C (1)				V_{CBO}
2N3838 (2)		2N4854 (3)		2N3838 (4)		2N4854 (5)		
One Transistor	Total Device	One Transistor	Total Device	One Transistor	Total Device	One Transistor	Total Device	
W	W	W	W	W	W	W	W	V dc
0.25	0.35	0.30	0.60	0.7	1.4	1.0	2.0	60

V_{EBO}	V_{CEO}	V_{1C-2C}	Lead to case voltage	I_C	T_J	T_{STG}
V dc	V dc	V dc	V dc	MA dc	°C	°C
5	40	±120	±120	600	200	-55 to +200

(1) T_C rating do not apply to surface mount devices (2N4854U).
(2) For T_A > +25°C, derate linearly 1.43 mW/°C one transistor, 2.00 mW/°C both transistors.
(3) For T_A > +25°C, derate linearly 1.71 mW/°C one transistor, 3.43 mW/°C both transistors.
(4) For T_C > +25°C, derate linearly 4.0 mW/°C one transistor, 8.0 mW/°C both transistors.
(5) For T_C > +25°C, derate linearly 5.71 mW/°C one transistor, 11.43 mW/°C both transistors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Primary electrical matching characteristics of each individual section. Characteristics apply to all case outlines.

	h_{FE5} (1) V_{CE} = 10 V dc I_C = 150 mA dc	h_{FE3} (1) V_{CE} = 10 V dc I_C = 10 mA dc	h_{FE2} V_{CE} = 10 V dc I_C = 100 µA dc	$V_{CE(sat)}$ (1) I_B = 15 mA dc I_C = 150 mA dc	$V_{BE(sat)}$ (1) I_B = 15 mA dc1.5 I_C = 150 mA dc
				V dc	V dc
MIN	100	75	35		0.80
MAX	300			0.4	1.25

	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 20 mA dc f = 100 MHz	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CEO} (2) I_C(on) = 600 mA dc I_B(on) = 120 mA dc I_B(off) = 0	$t_{(on)}$ See figure 4	$t_{(off)}$ See figure 5	$t_{on} + t_{off}$ See figure 6
		pF	V dc	ns	ns	ns
MIN	2		40			
MAX	8	8		45	300	18

(1) Pulsed (see 4.5.1).
(2) See 4.5.2.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

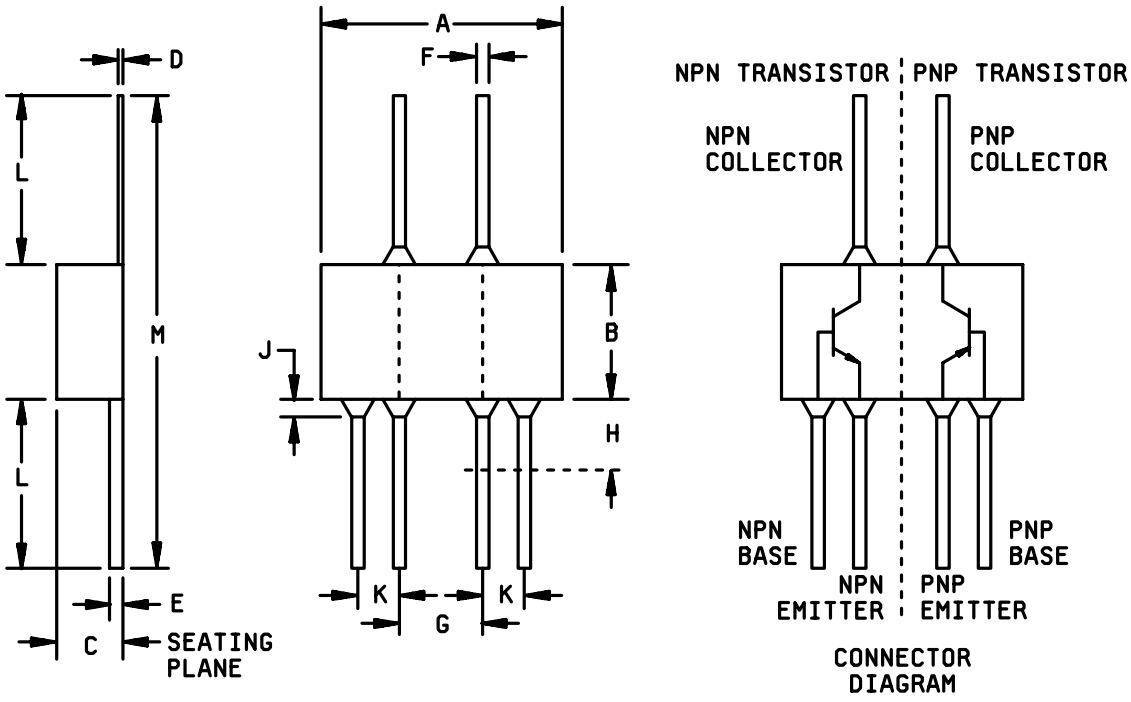
DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.240	.290	6.10	7.37	
B	.115	.160	2.92	4.06	
C	.030	.080	0.76	2.03	
D	.003	.006	0.08	0.15	4
E	.005	.035	0.13	0.89	
F	.010	.019	0.25	0.48	4, 6

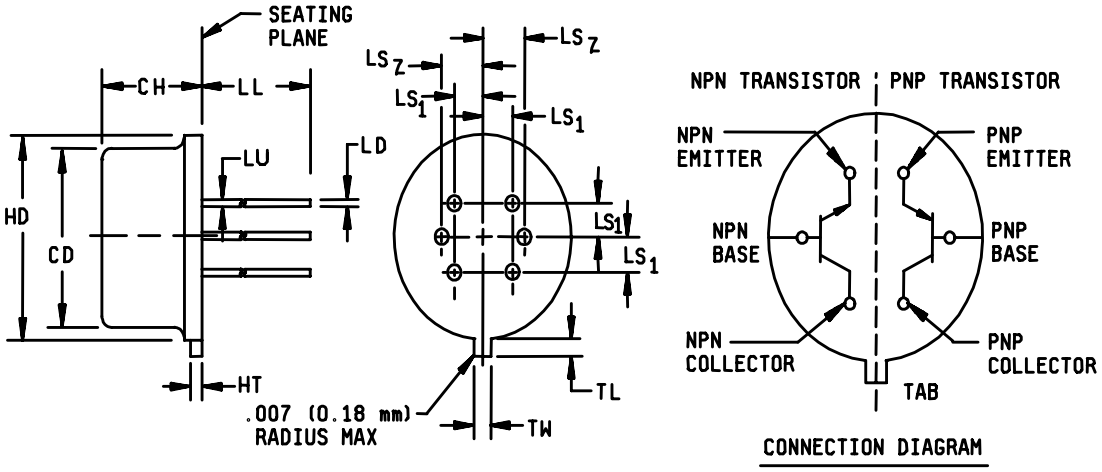
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
G	.100 TP		2.54 TP		6, 7
H		.050		1.27	
J		.015		0.38	5
K	.050 TP		1.27 TP		6, 7
L	.070	.250	1.78	6.35	3, 4
M	.260	.650	6.60	16.51	

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Maximum limit of this dimension does not apply to device supplied in a carrier.
4. All six leads.
5. Lead dimensions are uncontrolled in this zone.
6. Dimensions "F", "G", and "K" to be measured in zone "H".
7. Leads within .005 inch (0.13 mm) total of true position (TP) at "H" with maximum material condition.

FIGURE 1. Physical dimensions of transistor type 2N3838 (all quality levels).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.140	.260	3.56	6.60	
HD	.335	.370	8.51	9.40	
HT	.009	.125	0.23	3.18	
LD	.016	.021	0.41	0.53	3, 7
LL	.500	1.750	12.70	44.45	7

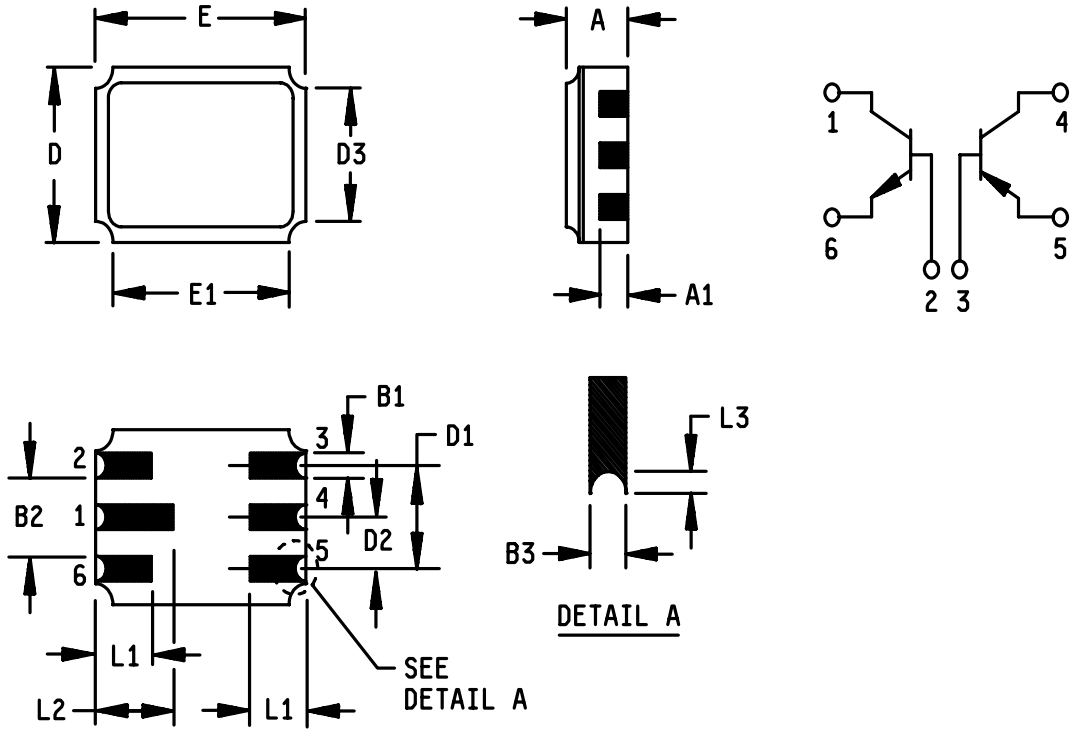
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
LS1	.0707 Nom.		1.796 Nom.		5
LS2	.1000 Nom.		2.540 Nom.		5
LU	.016	.019	0.41	0.48	4, 7
TL	.029	.045	0.74	1.14	6
TW	.028	.034	0.71	0.86	

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. When measured in a gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
6. Measured from the maximum diameter of the actual device.
7. All six leads.

FIGURE 2. Physical dimensions of transistor type 2N4854 (all quality levels).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.058	.100	1.47	2.54
A_1	.026	.039	0.66	0.99
B_1	.022	.028	0.56	0.71
B_2	.072 Ref.		1.83 Ref.	
B_3	.006	.022	0.15	0.56
D	.165	.175	4.19	4.45
D_1	.095	.105	2.41	2.67

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
D_2	.045	.055	1.14	1.40
D_3		.175		4.45
E	.240	.250	6.10	6.35
E_1		.250		6.35
L_1	.060	.070	1.52	1.78
L_2	.082	.098	2.08	2.49
L_3	.003	.007	0.08	0.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15 mm) for solder dipped leadless chip carriers.

FIGURE 3. Physical dimensions of transistor type 2N4854U.

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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.2).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

V_{1C-2C}.....Voltage difference between the collector of transistor one and that of transistor two.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (flat pack), 2 (similar to T0-77) and 3 (surface mount) herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.4, 1.5, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)
9	Not applicable
10	48 hours minimum
11	I_{CBO2} and h_{FE5}
12	Burn-in (see 4.3.1) 80 hours minimum
13	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 25 nA dc; whichever is greater for the 2N3838. ΔI_{CBO2} = 100 percent of initial value or 5 nA dc; whichever is greater for the 2N4854, 2N4854U. Δh_{FE5} = 15 percent of initial value.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); P_T = 1.0 W:

2N3838: P_T = 175 mW each transistor (350 mw total device).
2N4854: P_T = 300 mW each transistor (600 mW total device).
2N4854U: P_T = 300 mW each transistor (600 mW total device).

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

* 4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750.

a. I_H forward heating current ..... 200 mA.minimum.

b. t_H heating time ..... 25 -30 ms.

c. I_M measurement current ..... 5 mA.

d. t_{MD} measurement delay time ..... 60 µs maximum.

e. V_{CE} collector to emitter ..... 10 V.

The maximum limit for $Z_{\theta JX}$ under these conditions are $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in 4.4.2.1 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.1 and shall be in accordance with group A, subgroup 2 herein.

* 4.4.2.1 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 -30 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hours for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new "assembly lot" option is exercised, the failed assembly lot shall be scrapped.

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* 4.4.2.2 Group B sample selection. Samples selected for group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and 4.4.3.1 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) requirements shall be in accordance with table I, group A, subgroup 2 herein.

* 4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E, not applicable to U.
C6	1026	1,000 hours at V_{CB} =30 V dc; 2N4854, 2N4854U: P_T = 300 mW (one section); P_T = 600 mW (both sections); 2N3838: P_T = 175 mW each transistor (350 mW total device); at T_A = +30°C ±5°C. No heat sink or forced-air cooling on the device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Testing of units. All specified electrical tests, including end-point tests, shall be performed equally on both sections of the transistor types covered herein, except where the electrical characteristic being evaluated applies to the transistor as a device entity.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination	2071					
*Solderability 3/	2026	n = 15 leads, c = 0				
*Resistance to solvents 3/, 4/	1022	n = 15 devices, c = 0				
* Temp cycling 3/	1051	Test condition C, 25 cycles n = 22 devices, c = 0				
* Heremetic seal Fine leak Gross leak	1071	n = 22 devices, c = 0				
*Electrical measurements Bond strength 3/	2037	Group A, subgroup 2 Precondition T_A = +250°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		10	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}			
2N3838 2N4854 2N4854U					50 10 10	nA dc nA dc nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 3 V dc	I_{EBO2}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 µA dc	h_{FE2}	35		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE3}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	75		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc pulsed (see 4.5.1)	h_{FE5}	100	300	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 300 mA dc; pulsed (see 4.5.1)	h_{FE6}	35		
Base emitter saturation voltage	3066	Test condition A; I_B = 15 mA dc; I_C = 150 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$	0.80	1.25	V dc
Collector to emitter saturation voltage	3071	I_B = 15 mA dc; I_C = 150 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		0.40	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
* Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO3}		10	µA dc
Low temperature operation:	3076	T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE7}	12		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	60	300	
Small-signal common emitter input impedance	3201	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{ie}	1.5	9	kΩ
Small-signal common emitter output admittance	3216	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{oe}		50	µhmo
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2	10	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Noise figure	3246	V_{CE} = 10 V dc; I_C = 100 µA dc; f = 1 kHz; R_G = 1 kΩ	NF		8	dB
Turn-on time		See figure 4	t_{on}		45	ns
Turn-off time		See figure 5	t_{off}		300	ns

See footnotes at end of table.

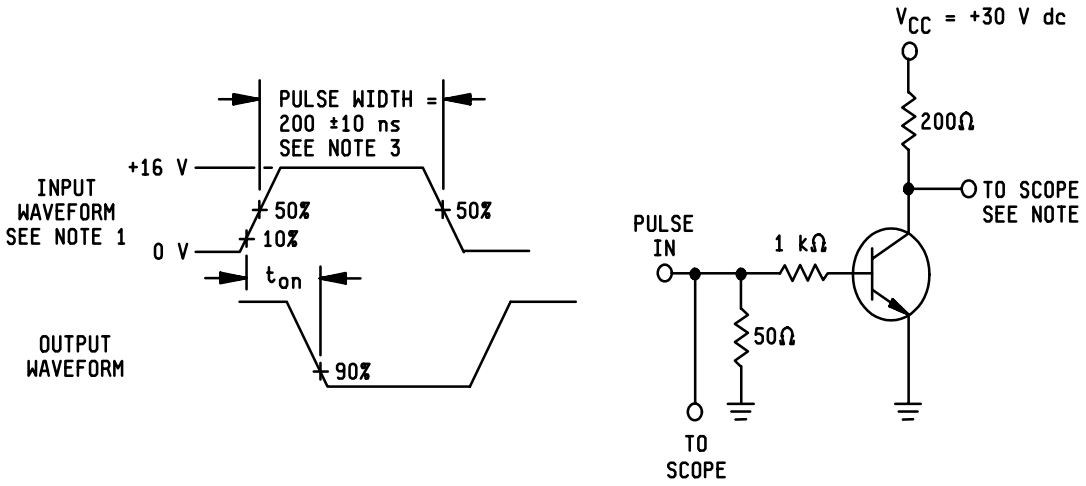
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TABLE I. Group A inspection - Continued.

	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Pulse response		See figure 6	$t_{on} + t_{off}$		18	ns
Collector emitter nonlatching voltage		See figure 7	V_{CEO}	40		V dc
Subgroups 5, 6, and 7						
Not applicable						

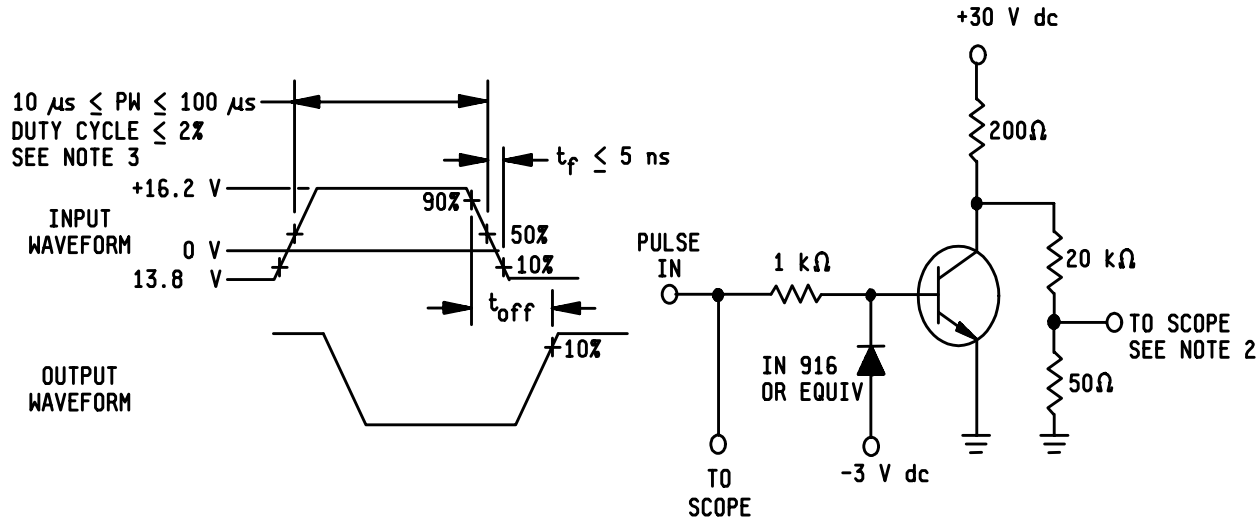
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ 0.2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

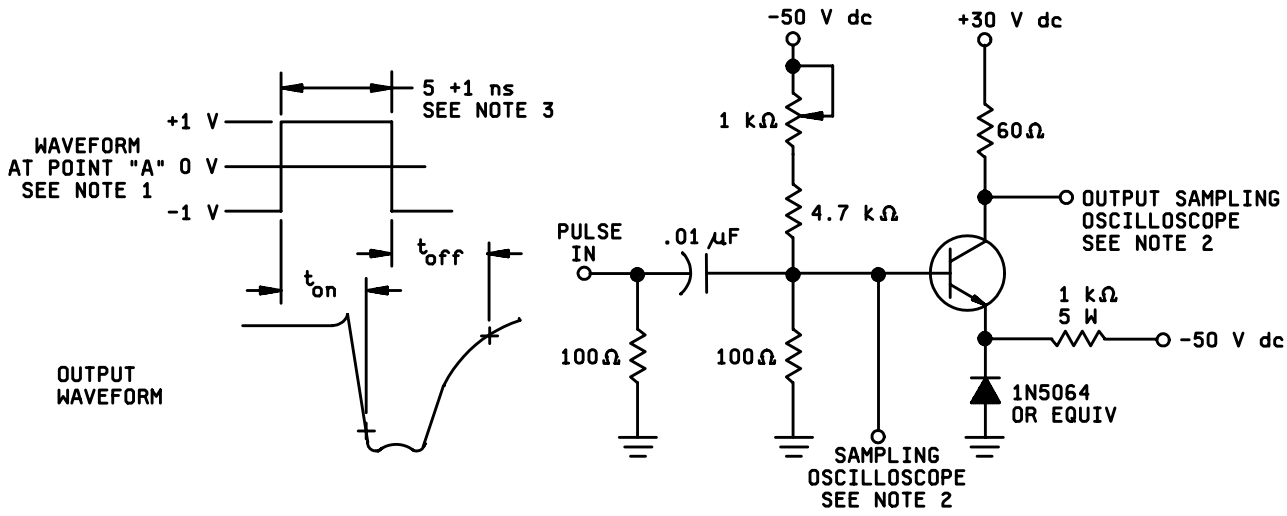
FIGURE 4. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ 0.2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

FIGURE 5. Saturated turn-off switching time test circuit.

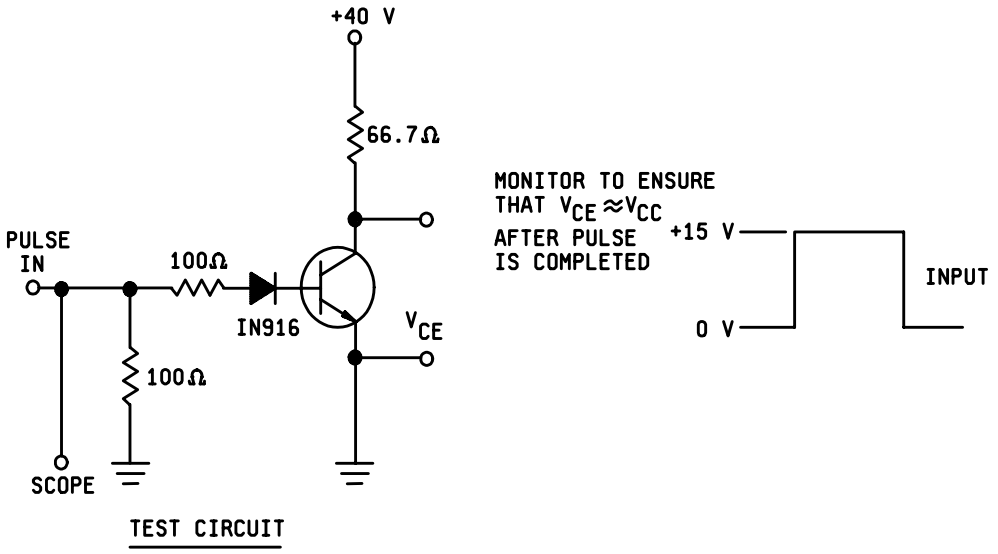
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ 0.2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

FIGURE 6. Nonsaturated switching time test circuit.



NOTE: The input waveform has the following characteristics: PW ≤ 10 µs, duty cycle ≤ 2 percent.

FIGURE 7. Collector emitter nonlatching voltage test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Packaging requirements (see 5.1).

d. Lead finish (see 3.4.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - CR
Navy - NW
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - MC

Preparing activity:
DLA - CC

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/421F	2. DOCUMENT DATE 25 August 2001

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, DUAL TRANSISTOR, UNITIZED, NPN/PNP, COMPLEMENTARY, SILICON, TYPES 2N3838, 2N4854 AND 2N4854U, JAN, JANTX, AND JANTXV.

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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